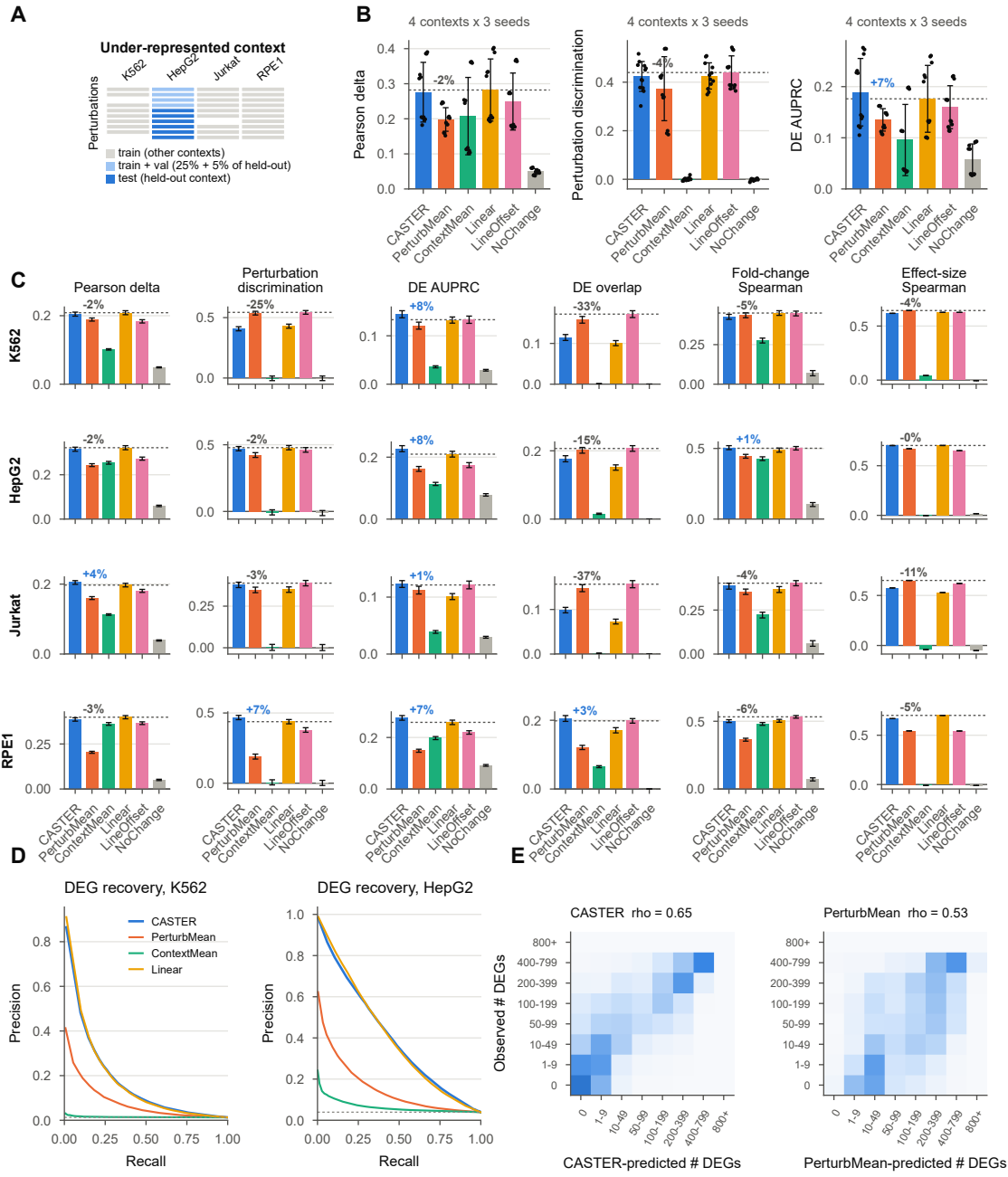

A
Under-represented context
K562
HepG2
Jurkat
RPE1
Perturbations
train (other contexts)
train + val (25% + 5% of held-out)
test (held-out context)
B
4 contexts x 3 seeds
Pearson delta
-2%
Perturbation discrimination
-4%
DE AUPRC
+7%
CASTER
PerturbMean
ContextMean
Linear
LineOffset
NoChange
C
Pearson delta
Perturbation discrimination
DE AUPRC
DE overlap
Fold-change Spearman
Effect-size Spearman
K562
HepG2
Jurkat
RPE1
-2%
-25%
+8%
-33%
-5%
-4%
-2%
-2%
+8%
-15%
+1%
-0%
+4%
-3%
+1%
-37%
-4%
-11%
-3%
+7%
+7%
+3%
-6%
-5%
D
DEG recovery, K562
DEG recovery, HepG2
CASTER
PerturbMean
ContextMean
Linear
Precision
Recall
E
CASTER rho = 0.65
PerturbMean rho = 0.53
Observed # DEGs
CASTER-predicted # DEGs
PerturbMean-predicted # DEGs
0
1-9
10-49
50-99
100-199
200-399
400-799
800+